CODA: The Calculus of Obstruction, Descent, and Authority

A unified account of witnessed difference, finite repair, representation, and scoped evidence

Author: Jeremy H. Carroll

Version: 2.0.1 — consolidated research manuscript

Date: 2026-09-28

Manuscript license: Creative Commons Attribution 4.0 International

Abstract

CODA studies what a declared representation can preserve, what its admitted repairs can change, and what evidence supports a subsequent decision. This paper consolidates the design, foundational manuscript, discipline statement, and recovery records in one exposition. Its starting point is a witnessed irreflexive relation; observation, algebra, repair families, costs, and authorization are additional data. The mathematical core consists of an image-factorization criterion, collision obstructions, laws for scoped repair menus, a finite-dimensional repair quotient, and elementary transport and gluing results. A common interface relates finite obstruction calculus (FOC), finite descent authority (FDA), re-presentation (MUR and OR), finite shadows (CS), continuum models (Q), their bridge (CS_Q), and typed residuals (A). GTLA supplies the proposed architecture for keeping evidence and grants separate. Exact examples exhibit the three FOC grades, local solutions without a global section, and identical finite samples with different integrals. An explicit counterexample prevents promotion of the inherited point-span construction to a faithful groupoid embedding. The accompanying ledger distinguishes local Lean results, prose proofs, finite calculations, policies, historical reports, and open targets. Consolidation provides one paper and a replayable finite companion; it does not settle the research programme's open foundation, continuum, or authority questions.

1. The question and the scope of the answer

Suppose an investigator records a state, observes only part of it, and tries to repair a failure. Three questions arise. Does the observation retain the information required by the query? Do the available repairs actually remove the failure? Does the resulting evidence satisfy the conditions of a separately supplied permission to act? A correct answer to one question can leave the others unresolved. CODA makes the objects and assumptions of each question explicit.

Naming convention (CODA-19). CODA means *Calculus of Obstruction, Descent, and Authority*, following the author-fixed expansion in the discipline source. Authorization names the act of granting or checking permission; it does not change the acronym. Earlier expansions and the alias RCOC remain withdrawn in the source history. Here *primitive difference* means the witnessed relation in §2; *residual* means data left by a stated repair or closure; *defect* means a stated diagnostic of such data. These objects are not interchangeable.

The central claim of this paper is a conditional one: once the state space, observation, query, and repair family are declared, explicit mathematical criteria can separate successful descent, a witnessed obstruction, and an unresolved search. This is useful even when the choice of those data remains contestable. The exposition does not infer its extra structure from a bare distinction.

Claims have stable identifiers. The claim ledger gives assumptions, supporting files, and the scope of mechanization. A prose proof is a proof of its written statement; a finite calculation checks its

declared instance; a Lean declaration verifies its formal statement and listed dependencies. None of these descriptions substitutes for another. The changelog records the substantive corrections.

1.1 Source basis

The source corpus contains two main expositions: a historical design paper and a shorter foundational manuscript. It also contains a discipline paper with code and Lean fragments, six recovery notes, a declaration-location scan, and fourteen diagrams. The synthesis uses the foundational manuscript for its mathematical order, the design for its operational interfaces, the discipline for its naming and withdrawal rules, and the recovery notes for the component relationships and unresolved evidence obligations.

Source scope (CODA-20). The source manifest verifies seventeen local historical files from baseline commit `d435c7f86d2896b1b738d45abd7f47a690dcd388`: fourteen diagrams and three byte-preserved discipline source transcripts in `docs/source_archive/`. Each entry records its Git blob, byte count, SHA-256, and immutable retrieval URL. The active code and Lean files are maintained derivatives, distinct from those originals. The other twenty-two tracked inputs, including the historical root manuscripts, ledgers, recovery notes, and scanner, have immutable external retrieval pointers in the same manifest. They are not claimed as local copies. Six formerly reported ignored build logs cannot be recovered from that Git commit and are excluded from the present custody claim.

The active exposition is `coda.md`; its current code, formal sources, and notebooks have separate release checksums. Historical file-and-line citations retain their original context and do not certify current builds.

2. Witnessed difference and admitted observation

2.1 A distinction with an actual witness

A **witnessed difference** is a tuple

$$W = (X, D, a, b, w), \qquad D \subseteq X \times X, \quad (\forall x \in X)\ \neg D(x, x), \quad w : D(a, b).$$

The relation is irreflexive and a related pair is supplied. Symmetry, transitivity, decidability, finiteness, and a numerical magnitude are absent from this definition. Ordinary equality in the ambient mathematics is already available; a derived observational equivalence below is not a construction of every possible notion of identity.

Proposition 2.1 (CODA-01; inhabited distinction). The supplied elements satisfy $a \neq b$.

Proof. Substituting $a = b$ into $D(a, b)$ gives $D(a, a)$, contradicting irreflexivity. The witness also rules out an empty carrier. ■

A detector is a different signature. It consists of $\delta : X \times X \to V$ and a designated 0_V. If $\delta(x, x) = 0_V$ for every x, and $\delta(a, b) \neq 0_V$, then $D_\delta(x, y)$ defined by $\delta(x, y) \neq 0_V$ supplies a witnessed irreflexive relation. The proof of $a \neq b$ is the same substitution argument. Symmetry requires another law. A quotient by $\delta(x, y) = 0_V$ requires that zero relation to be an equivalence relation. Neither a zero diagonal nor a nonzero pair supplies all those laws.

This distinction reconciles two objects called "primitive difference" in the sources: the detector-and-quotient primer and the witnessed relation. The former is an admitted evaluation model. The latter is the starting signature adopted here. Their conditional bridge does not identify them.

2.2 Observation and the query it must preserve

Add an observation $p : X \to P$. Define $x \sim_p y$ when $p(x) = p(y)$. Equality in P makes this relation reflexive, symmetric, and transitive. Those laws come from the observation model, not from the irreflexive relation D.

Add a query $q : X \to Y$. Let $I = p(X)$ be the **realized image**, considered as a subset of P. Query descent asks for a map $\bar{q} : I \to Y$ satisfying $\bar{q}(p(x)) = q(x)$.

Theorem 2.2 (CODA-02; descent on the image). A unique image factor $\bar{q}$ exists if and only if the query is constant on every observation fiber:

$$p(x) = p(y) \quad \Longrightarrow \quad q(x) = q(y).$$

Proof. A factor sends equal observations to equal answers. Conversely, for each realized z, choose an x with $p(x) = z$ and put $\bar{q}(z) = q(x)$. Fiber constancy makes this definition independent of the representative. Any other factor agrees at z, since z has a representative. If X is empty, so is its image, and the empty function is the unique factor. ■

The local Lean implementation uses classical selection for the image factor; its uniqueness proof reports `Quot.sound`. These are explicit dependencies, not a claim of an executable inverse on an arbitrary infinite space. On a supplied finite domain the Python routine constructs the factor by enumerating fibers and uses no such infinite selection procedure.

A map on all of P is a stronger request. The theorem ensures no value off the image. If $X = \emptyset$, P is a singleton, and $Y = \emptyset$, the image factor exists but no map $P \to Y$ does. Even when an extension exists, off-image values need not be unique.

2.3 Collisions and the limits of post-processing

A **collision witness** is a pair x, y with the same observation and different query answers. On $X = \{0,1\}^2$, the observation $p(u,v) = u$ cannot recover $q(u,v) = v$, as $(0,0)$ and $(0,1)$ witness. It does recover $q(u,v) = 2u$. Adequacy is query-relative.

Proposition 2.3 (CODA-03; collision persistence and repair). A collision rules out a factor through p and through every coarsening $c \circ p$. Any new observation p' through which the same query factors must separate that pair.

Proof. Equality $p(x) = p(y)$ remains equality after applying c. A factor would then imply $q(x) = q(y)$, a contradiction. Applying the same argument to p' proves the second assertion. ■

The last conclusion is necessary, not sufficient: separating one exhibited pair can leave another collision. Bundling an informative additional observation can help; repeating or relabeling the same collapsed observation cannot. A detector that displays the same value for every state is adequate only for queries constant on that whole domain.

3. Repair families, search budgets, and extension

A repair problem supplies a set A of admitted moves and a predicate $R(a)$ saying that move a meets the protected requirement. A protected requirement might preserve a query, maintain a distinction, or satisfy a constraint. Its choice is part of the problem. Define

$$\text{Exhausted}(A, R) \quad :\Longleftrightarrow \quad (\forall a \in A)\ \neg R(a).$$

An obstruction is this scoped assertion or a proved criterion for it. A search log that stops after checking part of A does not establish it. An empty menu is vacuously exhausted; this is not evidence about any larger class of possible repairs.

Proposition 3.1 (CODA-04; exhaustion and ties). A working member of A refutes exhaustion. If $B \subseteq A$, exhaustion of A implies exhaustion of B; the reverse implication fails. If a cost $c : A \to V$ takes values in a preorder and $c(a) = c(b)$, every attained minimum property of a is also a minimum property of b.

Proof. Instantiate the exhaustion assertion at the working member. Restricting the universal assertion proves the submenu implication. For its failed converse, take $A = \{0, 1\}$, $B = \{0\}$, and $R(a)$ meaning $a = 1$. For ties, substitute $c(b) = c(a)$ into $c(a) \leq c(x)$ for every candidate x. ■

No minimum-existence theorem follows merely from a preorder. A finite nonempty successful menu with rational costs does attain its least cost; the companion enumerator returns **all** tied minimizers. During a partial search it may report working witnesses, but withholds a minimum claim because an unchecked candidate may be cheaper.

The source phrase "remove the obstruction or construct structure" is used here as a workflow rule. A **repair** uses already declared operations. An **extension** changes the observation, carrier, operations, or requirements and therefore creates a new problem whose relationship to the old one must be stated. Failure of all current repairs does not prove that a successful extension exists or select one. Refusal records a reason to stop an attempt. It can coexist with a residual and an open extension question.

The cost measure and any choice among tied candidates are separate declarations. Ties may be identified only under an explicitly justified equivalence that preserves the relevant success and cost properties. Counting one class after such a quotient is a statement about that quotient.

4. Finite obstruction calculus: a declared algebraic extension

4.1 The packet and its quotient values

FOC adds finite-dimensional real vector spaces and linear maps

$$C^0 \xrightarrow{d_0} C^1 \xrightarrow{d_1} C^2, \qquad U^2 \xrightarrow{a_2} C^2, \qquad d_1 d_0 = 0.$$

Choose coordinates and strictly positive weights w_i, defining $\|v\|_{1,w} = \sum_i w_i |v_i|$. The full declared linear repair subspaces are $M_1 = \operatorname{im} d_0$ and $M_2 = \operatorname{im} a_2$. "Admitted" here means belonging to the mathematical repair class; it does not mean institutionally authorized.

For $x \in C^1$, define

$$\Phi_1([x]) = \min_f \|x + d_0 f\|_{1,w_1}, \qquad \Gamma_2([d_1 x]) = \min_u \|d_1 x + a_2 u\|_{1,w_2}.$$

Theorem 4.1 (CODA-05; quotient diagnostics and trichotomy). These minima are attained, depend only on the displayed quotient classes, and vanish exactly when their arguments belong to the respective repair subspaces. Exactly one of the following grades holds:

Grade	Condition	Meaning within this packet
`ExactRepair`	$\Phi_1 = 0$	A declared stage-one repair removes the vector.

Grade	Condition	Meaning within this packet
`CoherentResidual`	$\Phi_1 > 0,\ \Gamma_2 = 0$	The first class remains; its next defect is stage-two repairable.
`NonclosedDefect`	$\Phi_1 > 0,\ \Gamma_2 > 0$	Both stated repair quotients detect a remaining class.

Proof. Each repair set is an affine translate of a finite-dimensional linear subspace and is closed. A sequence whose norms approach the infimum can be chosen bounded. In finite dimension it has a convergent subsequence; the limit remains in the affine subspace and attains the infimum. Strictly positive weights make zero norm equivalent to the zero vector. Thus zero minimum is equivalent to membership in the repair subspace. Translating a representative by an element of that subspace leaves the affine repair set unchanged. Finally, $d_1 d_0 = 0$ ensures that changing the first representative does not change $d_1 x$, and that $\Phi_1 = 0$ implies $\Gamma_2 = 0$. Nonnegativity then gives the three disjoint cases. ■

This theorem has a prose proof here. The companion code implements the explicit packet below, not a general optimizer, and the local Lean library does not mechanize the finite-dimensional compactness argument. With zero weights, a nonzero vector can have zero score. With nonlinear, incompletely enumerated, or nonclosed repair sets, the proof needs different hypotheses. A failed packet validation precedes classification.

4.2 A certificate that can be checked independently

Proposition 4.2 (CODA-06; a matching dual bound). Suppose $r = x + d_0 f$, a covector y satisfies $y d_0 = 0$ and $|y_i| \le w_i$, and $y(x) = \|r\|_{1,w}$. Then f is an optimal repair and that common value equals $\Phi_1([x])$.

Proof. For every g, annihilation gives $y(x + d_0 g) = y(x)$, while the coordinate bounds give $y(x + d_0 g) \le \|x + d_0 g\|_{1,w}$. The supplied repair attains this lower bound. ■

The proposition checks a supplied certificate; it does not assert that a particular solver always finds one. A stored certificate should identify the packet, weights, and query it concerns. The example JSON records its exact input, and the package manifest binds its bytes to the implementation.

Take $C^1 = \mathbb{R}^3$, $d_0(t) = (t, 0, 0)$, $d_1(a, b, c) = c$, and $a_2 = 0$, with unit weights. Then

$$f = -a, \quad r = (0, b, c), \quad \Phi_1 = |b| + |c|, \quad \Gamma_2 = |c|, \quad y = (0, \operatorname{sgn} b, \operatorname{sgn} c).$$

The covector supplies the matching lower bound even though infinitely many real repairs were available. The exact rational examples are:

Input	Repair	Residual	Φ_1	Γ_2	Grade
$(2, 0, 0)$	-2	$(0, 0, 0)$	0	0	`ExactRepair`
$(2, 3, 0)$	-2	$(0, 3, 0)$	3	0	`CoherentResidual`
$(2, 3, 5)$	-2	$(0, 3, 5)$	8	5	`NonclosedDefect`

"Coherent" describes the next quotient in this packet. It does not establish stability under every extension, a global mathematical object, or authority.

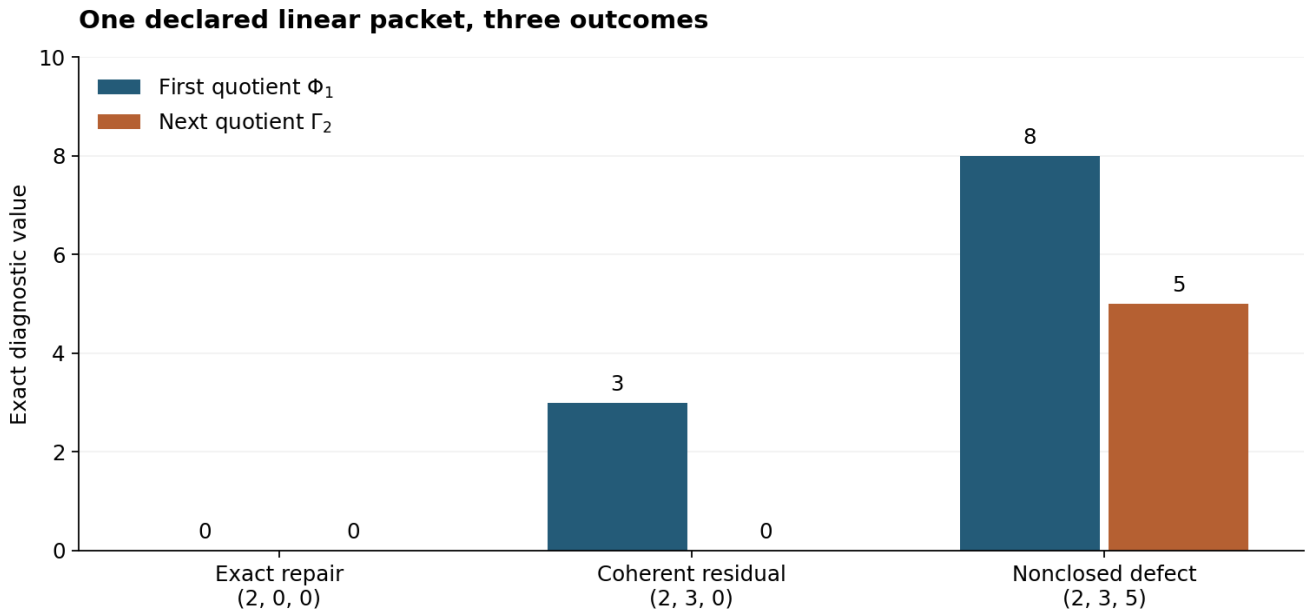


Figure 1: Two diagnostics distinguish three outcomes under the same declared repair maps. The numbers are finite example values, not measured physical quantities.

5. Representation transport and local-to-global obstruction

5.1 MUR: what a transport certificate establishes

A representation transport consists of maps $F : X \to X'$ and $f : P \to P'$, observations p, p', and a commuting-square obligation

$$p' \circ F = f \circ p.$$

Proposition 5.1 (CODA-07; transported query factor). If $q' = \bar{q}' \circ p'$, then $q' \circ F = \bar{q}' \circ f \circ p$.

Proof. Substitute the square identity into $q'(F(x)) = \bar{q}'(p'(F(x)))$. ■

This statement transfers a particular query. It does not make F injective, supply an inverse, preserve every other query, or supply a grant. For FOC packets, compatible linear transports satisfy $T_1 d_0 = d_0' T_0$, $T_2 d_1 = d_1' T_1$, and $T_2 a_2 = a_2' S_2$. These laws preserve the relevant algebraic relations. Quantitative norm preservation additionally needs a bound or isometry law.

MUR contributes a **loss ledger**: which protected distinctions were erased, identified, approximated, or left unresolved by a transport. These are reports about the named map. A field called "lossless" cannot replace a proof that a protected query is preserved. Subtraction of two transported objects is meaningful only after an additive comparison object has been chosen; a general commuting-square failure is first a failed equality.

5.2 OR: local solvability is insufficient for a global section

Comparison among several presentations adds overlap and compatibility data. Let three vertices carry values $s_0, s_1, s_2 \in \mathbb{F}_2$. Impose

$$s_0 + s_1 = 0, \qquad s_1 + s_2 = 0, \qquad s_2 + s_0 = 1.$$

Proposition 5.2 (CODA-08; an explicit gluing obstruction). Each edge equation has solutions, but the three equations have no common solution.

Proof. An equality edge admits $(0,0)$; the unequal edge admits $(0,1)$. The first two global equations imply $s_0 = s_1 = s_2$, contradicting the third. Equivalently, summing the three equations in $\mathbb{F}_2$ gives $0 = 1$. ■

There are four assignments on the three-vertex carrier satisfying any single edge equation and zero satisfying all three. The local Lean proof uses Boolean equality and inequality for the same pattern. This is a local-solvability counterexample: it does not supply a compatible family of local sections and then refute a sheaf gluing axiom. The missing condition is precisely compatible choices on shared vertices.

Averaging the three edge verdicts does not repair the contradictory system. One must change a constraint, extend the representation with an explicit translation, or retain the obstruction. Each action changes the stated problem and belongs in the loss and change records.

6. Finite shadows, continuum models, and query descent

6.1 The roles of CS, Q, and CS_Q

A finite shadow is a declared observation $p_\lambda : X \to P_\lambda$. A family may have refinement maps and compatibility laws. Calling its index set a scale does not itself supply an order, a topology, cofinality, an inverse limit, or a convergence theorem. Q concerns continuum models when their required metric, topological, analytic, or completion data are admitted. CS_Q asks whether a particular continuum query factors through the relevant shadow or family of shadows.

The image-factor theorem answers the exact descent question. It does not construct the continuum domain, prove continuity, or turn infinitely many conditions into a finite verifier. Approximate descent instead needs a declared error notion and a bound; exact and approximate questions should not share an unqualified success label.

6.2 Finite samples do not determine an arbitrary continuous integral

Proposition 6.1 (CODA-09; the sampling collision). On a nondegenerate compact real interval, a finite set of point samples does not determine the integral of every continuous function.

Proof. For sample nodes $x_1, \ldots, x_k$, set $h(x) = \prod_i (x - x_i)^2$; for no nodes take $h = 1$. This polynomial vanishes at every node, is nonnegative everywhere, and is positive on some subinterval because a finite set does not fill the interval. Its integral is strictly positive. Thus f and $f + h$ have identical samples but different integrals, giving a collision for the integral query. ■

The proof is analytic prose, not a theorem in the local Lean package. An exact polynomial instance is replayed in Python. On $[0, 10]$, use samples at $0, 5, 10$, put $u = x/10$, and choose

$$f(x) = x, \qquad h(x) = [u(u - \tfrac{1}{2})(u - 1)]^2.$$

Direct polynomial integration gives $\int_0^{10} h(x)\,dx = 1/84$. The common samples are $(0, 5, 10)$, but the integrals are 50 and $50 + 1/84 = 4201/84$. If the admissible class is instead affine functions, distinct endpoint samples determine the function and its integral. That is a narrower domain with an explicit reconstruction law.

6.3 A worked calculation with all boundaries visible

Proposition 6.2 (CODA-10; affine quadrature). For $f(x) = x$ on $[0, 10]$, an equal partition with $N \geq 1$ panels gives

$$L_N = 50 - \frac{50}{N}, \qquad R_N = 50 + \frac{50}{N}, \qquad T_N = \frac{L_N + R_N}{2} = 50 = \int_0^{10} x\, dx.$$

Proof. With $\Delta x = 10/N$, the left sum is $(100/N^2)\sum_{i=0}^{N-1} i$, and the right sum replaces the index range by $1, \ldots, N$. The two arithmetic-sum formulas give the displayed values. Their average is 50; direct integration gives the same value. ■

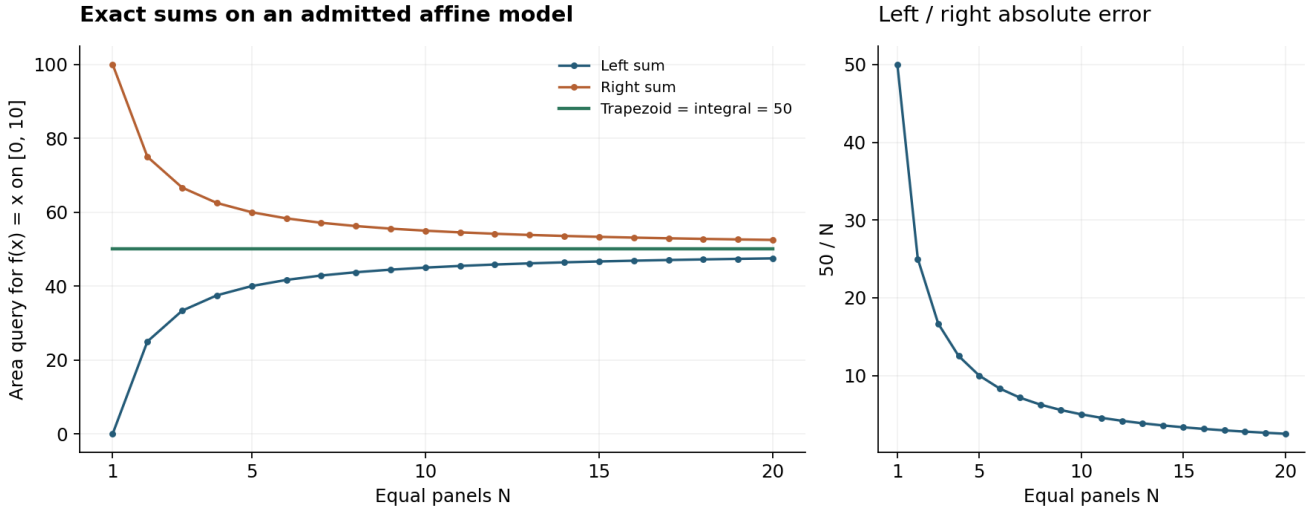


Figure 2: The trapezoid value is exact for the admitted affine function. The plotted finite panel counts and errors do not reconstruct arbitrary continuous functions or establish a new continuum theory.

At $N = 10$, the three values are 45, 55, and 50. Their relationship has a proof; it is not a vote for the middle answer. The scalar error $50/N$ is a diagnostic relative to this quadrature and this query. If a full repair family contains the exact trapezoid replacement, the residual after optimal repair is zero. Calling the pre-repair quadrature error a surviving FOC obstruction would be incorrect.

The continuum interval and ordinary integral were admitted before this calculation. The finite calculation evaluates a query about them. It neither derives their existence from witnessed difference nor authorizes an action.

7. FDA and GTLA: evidence, standing, and permission

FDA organizes the movement from a proposal to a finite packet, verifier result, and scoped decision grade. GTLA is used here for the architecture that records how evidence and separately supplied grants relate. A proposed admission record identifies the proposal, its scope, the supporting evidence, and its grant. Verifier soundness and the grantor's standing are assumptions whose justification must be supplied outside a mere status flag.

Policy and proposition 7.1 (CODA-11). In the declared logical model $\text{Standing}(E, G) := E \wedge G$, standing implies a grant, whereas evidence alone does not imply standing.

Proof. A conjunction supplies its second component. The valuation $E = \text{True}$, $G = \text{False}$ has evidence and lacks standing. ■

The theorem is elementary because the policy is explicit. It proves a property of that model. It does not authenticate a grantor, enforce roles, provide a cryptographic signature, or prove that every possible institution must use this policy. An authorization system may use automated rules, but the rules and their authority must then be admitted and justified as data of that system. Mathematical evaluation does not supply them merely by returning a low defect.

The historical sources sometimes put "authority" inside a terminal type or inside a conjunction of successful checks. Here a terminal reports the evaluation outcome; an admission record separately records a grant. Merely renaming a Boolean conjunction as a dependent record does not authenticate its contents. Conversely, a typed residual and a valid grant for some other scope may coexist on one object. There is no assertion that every residual forbids every kind of permission.

Typical outcomes include a successful repair witness, a negative result over the declared menu, a retained tie, `BudgetUnknown`, a typed refusal, and an open extension obligation. A refusal can accompany a calculation; it is not represented by a large numerical defect. "No rule in this menu" does not become "false globally." A receipt supports a statement about a particular run only to the extent that its inputs, computation, and verifier are actually checked.

8. Typed residuals and the unified research architecture

8.1 A: the remainder of a declared closure attempt

Let a closure discipline Λ specify an attempted construction, its comparison, and the target class of acceptable results. A residual is what remains under that declared attempt. A categorical cofiber residual requires a category and a cofiber construction; it is not another name for every numerical error. A decomposition of a residual requires actual decomposition data and a subsequent classification of its pieces.

The A sources contribute the rule that unsuccessful closure is recorded as absorption where proved, a typed remainder, a refusal, or an unresolved attempt. An unverified claim that every repair was tried is insufficient to call a remainder permanent. An observer that collapses the distinction of interest can also hide the remainder. A "surviving anomalon" therefore requires explicit closure, observer, and nonvanishing hypotheses. No existence axiom for anomalons is used by the local proofs in this package.

8.2 Eight components and one authority architecture

Definition 8.1 (CODA-13; organizational interfaces). The eight component names designate the responsibilities below. GTLA is the overarching authority architecture rather than an additional necessary mathematical layer. This arrangement is an editorial and research organization, not a theorem that exactly these components must exist.

Component	Admitted input	Main question or output	Boundary
FOC — Finite Obstruction Calculus	Complex, full repair subspaces, positive weights	Quotient class, diagnostic, checked certificate	Low defect does not supply permission.
FDA — Finite Descent Authority	Proposal, packet, verifier, grant conditions	Scoped evaluation and admission conditions	Soundness and grantor legitimacy remain assumptions.
MUR — Meta and Universal Re-Presentation	Representations and a declared family of maps	Commuting comparisons and loss records	Transport need not be invertible or norm preserving.

Component	Admitted input	Main question or output	Boundary
OR — Omniversal Re-Presentation	A declared diagram and compatibility data	Global section or gluing obstruction	Local solvability alone is insufficient.
CS — Continuum Shadow	Observation family and declared refinement laws	Query-relative finite information	A family name supplies no limit theorem.
Q — Continuum	Explicit analytic or completion data	Continuum model and scoped queries	No quantum or ontology theorem follows from the name.
CS_Q — Continuum Shadow–Continuum Bridge	Profile map and query	Factor on the realized image, or a collision	An image factor is not a total inverse.
A — Anomalon	Closure discipline, observer, typed remainder	Residual classification and next obligation	A remainder supplies neither its own origin nor authority.

The inherited words "Universal" and "Omniversal" in component names do not quantify over every representation or every world. Every theorem still names its actual domain. FOC's linear model and Q's analytic models are extensions of the foundational language; neither is inferred from it.

8.3 One operational loop, with its status stated

Declared workflow (CODA-14). A proposed run records the following stages:

1. Supply a proposal or datum and record its source.
2. Declare the state space, observation, protected query or constraint, repair menu, cost, budget, and verifier before the relevant evaluation.
3. Evaluate and return a witness, obstruction, tie, unresolved result, or refusal with its scope.
4. Seek or check a grant when an actual admission requires one under the chosen policy.
5. Apply the admitted change, replay its evidence, and record remaining obligations and losses.

Within this proposed interface, supply presents information without itself changing the admitted mathematical state. An admission is the designated state-changing operation. This is an interface choice here, not a claim that all software or reasoning changes state in this way.

Operation	Required record	What the record alone does not establish
Supply	Source and content	Admission
Admission	Named change, scope, grant when required	Grantor legitimacy
Observation	Domain, map, query	Adequacy for every query
Obstruction	Witness or quantified repair exclusion	Global impossibility
Menu	Enumerated domain or a justified quantifier	Exhaustion from a timeout
Cost	Measure and comparison rule	A uniquely preferred measure
Ties	All retained minimizers or justified classes	A selected representative
License	Grant object and scope	Permission outside that scope
Repair	Move and preserved requirements	Existence of an extension
Result	Typed payload and diagnostic	A grant

Operation	Required record	What the record alone does not establish
Terminal	Stop condition and extent of evaluation	Truth outside that extent
Refusal	Reason and affected operation	A numerical residual
Replay	Inputs, implementation, command, output	Semantic correctness by hashing alone
Leftover	Open question, blocker, proposed next test	A result already obtained
Order	Event sequence or commit relationship	A physical or intrinsic clock

A research run, a chain of runs, and a library entry may each be organized with these records. Their similarity is a design analogy. Encoding and replaying a real research run inside the sealed kernel remains a separate open research obligation.

8.4 The six foundational authority boundaries and invariant enforcements

To prevent epistemic conflation across proposal generation, evidence verification, and normative authorization, CODA integrates six foundational invariant boundaries formulated as declared policies (Papers 004, 005, 027, 028, 029, and 030).

Policy 8.3 (CODA-21; Coproduct Separation Invariant). In the declared evaluation-authority interface, the evidence type $\mathbb{E}$ and the authority terminal $\mathbb{T}$ are strictly separated into a coproduct:

$$\mathbb{T} := \mathbb{T}_{\mathsf{auth}} \amalg \mathbb{T}_{\mathrm{res}} \amalg \mathbb{F}_{\mathrm{term}},$$

where $\mathbb{T}_{\mathsf{auth}}$ denotes an authorized grant, $\mathbb{T}_{\mathrm{res}}$ denotes a typed residual with explicit unresolved debt, and $\mathbb{F}_{\mathrm{term}}$ denotes a terminal refusal. The declared policy requires guards appropriate to each branch. For admission into the authorizing terminal $\mathbb{T}_{\mathsf{auth}}$, all seven FDA guards must hold, including g_3 `zero_residual_mass` and g_7 `no_refusal`, with the guards derived from an evidence record and an external grant supplied separately (Paper 027). For non-authorizing terminals ($\mathbb{T}_{\mathrm{res}}$ and $\mathbb{F}_{\mathrm{term}}$), admission does not assert zero residual mass or absence of refusal; rather, $\mathbb{T}_{\mathrm{res}}$ records positive residual mass under valid determinism and structure preservation, while $\mathbb{F}_{\mathrm{term}}$ routes a terminal refusal upon detecting an obstruction or scope violation. The FDA guard identifiers are g_1 `accepted`, g_2 `deterministic`, g_3 `zero_residual_mass`, g_4 `skeleton_preserving`, g_5 `exhaustive`, g_6 `no_pending_lift`, and g_7 `no_refusal`. These names refer to `coda_fda_core.derive_guards` in the separate FDA package.

Policy 8.4 (CODA-22; Role Non-Impersonation Invariant). The proposal category $\mathcal{P}$, evaluation category $\mathcal{E}$, and authorization category $\mathcal{A}$ have disjoint morphism classes. In particular, no morphism in $\mathcal{P}$ or $\mathcal{E}$ can be cast or type-pushed into $\mathcal{A}$. The policy requires an automated solver, heuristic searcher, or reasoning system to act within $\mathcal{P}$ without issuing its own verification receipt or execution grant (Paper 004). This is a design requirement; this section supplies no proof that an implementation enforces it.

Policy 8.5 (CODA-23; Authority Gap Invariant). Epistemic verification does not entail normative authorization:

$$\mathrm{eval}(x) = \mathrm{valid} \;\not\Rightarrow\; \mathrm{grant}(x) = \mathrm{authorized}.$$

A vanishing mathematical residual ($\|\Omega\| = 0$) or valid dual witness certificate confirms mathematical consistency; it does not supply administrative or operational entitlement to execute state changes (Paper 005).

Policy 8.6 (CODA-24; Reasoning Authority Non-Generation Invariant). Under the assumption that each evaluation step preserves the evaluation domain ($\mathrm{step}(\mathcal{E}) \subseteq \mathcal{E}$), finite computation cycles, deductive steps, and heuristic tree searches cannot synthesize authority tokens:

$$\forall n \in \mathbb{N}, \quad \mathrm{step}^n(\mathcal{E}) \subseteq \mathcal{E}.$$

Every reasoning loop is **required** to run under a finite budget that can interrupt it ($T < T_{\max}$); a check made only between steps does not satisfy this (see change log regarding the 2026-09-23 withdrawal of implementation claims). If the budget is exhausted without terminal certificate generation, the system returns `BudgetUnknown` or a typed refusal, rather than an unearned pass (Paper 028).

Policy 8.7 (CODA-25; Anomalon Boundary Invariant). The primitive ontology of the system is partitioned into difference and structure:

$$\mathrm{Real}_0 = \mathrm{Diff} \cup \mathrm{Struct}.$$

The declared policy (Paper 029; see change log): a class whose exact ℓ^1 quotient distance (the FOC-T2 optimum, not a feasible dual pairing) is positive is retained, not smoothed, averaged or suppressed, in an append-only register whose fingerprint covers the shape-qualified defect data. `apps/coda` implements this with a JSONL file in which the first registration of a fingerprint wins. The file is persistent, not immutable: whoever controls the file system can delete it.

Policy 8.8 (CODA-26; Verification Limits Invariant). A finite formal verifier operating over finite states cannot decide arbitrary undecidable authorization envelopes, nor evaluate self-referential authorization envelopes lacking an admitted grounded carrier or stratified fixed point. Self-reference alone does not establish undecidability without a demonstrated diagonal obstruction. When evidence identifies such a scope limitation, the policy requires the system to halt closed into $\mathbb{F}_{\mathrm{term}}(\mathrm{refusal})$ with explicit scope documentation, rather than defaulting to an open execution license (Paper 030). The condition must be derived from the envelope; an unbound caller assertion does not establish it. This policy does not supply a general decision procedure for undecidable envelopes.

9. A failed construction and the faithful-functor target

The discipline source proposes a category of finite presentations modulo span isomorphism, an intensional trace category, an erasure functor, and a faithful functor from each finite groupoid. Its accompanying implementation does not establish that target.

Proposition 9.1 (CODA-12; the inherited point-span collision). The inherited construction is not faithful on the one-object groupoid $\mathbb{Z}_2$.

Proof. The distinct arrows e and g both map to the same span $1 \leftarrow 1 \rightarrow 1$, with the same leg maps and zero cost. Their images are equal even before passing to span-isomorphism classes. Thus the map on that hom-set is not injective. ■

The Python method `check_faithfulness()` checks uniqueness of dictionary keys, so it returns true for these two distinct names despite the collision of their represented values. The included audit runs the actual preserved source and records this contradiction. The inherited Lean construction also ignores its arrow argument in `groupoidArrowSpan`; its seven stated theorems concern costs and erasure, not injectivity. The new local Lean theorem proves the general obstruction for a constant map on two Boolean inputs, while the source audit binds that pattern to the Python example.

Open target (CODA-17). Construct the proposed presentation and trace categories, establish identity and associativity on the declared morphism classes, and prove a faithful functor with the advertised transport structure. A repair must distinguish parallel arrows in the represented hom-set; unique source

names are insufficient. Extensional cost must be well defined on the quotient, while intensional work belongs to traces. Inverse round trips can have identity as their erased morphism and positive declared trace work, but a cost calculation alone proves no functor law.

The counterexample refutes this candidate, not the existence of every possible faithful construction. The target remains open in this package. No subsumption of homotopy type theory, higher category theory, or any other foundation is asserted.

10. Evidence status, historical custody, and limitations

10.1 Current local evidence

The companion Python core implements finite image factorization, bounded menu evaluation, the coordinate FOC packet and an independent certificate checker, binary constraint enumeration, and exact quadrature and polynomial examples. It rejects floating-point and Boolean values at its rational arithmetic boundary. Its calculations exercise the written statements; exhaustive Boolean tests concern the enumerated domains.

The Lean root builds `Foundations.lean` and `Boundaries.lean` with Lean 4.33.0 and `Std` only. The twenty audited declarations cover witnessed distinction, image descent, collision laws, repair-menu scope, ties, a transport identity, the triangle example, and two elementary non-inference examples. General norm optimization, continuous integration, a faithful groupoid functor, the whole component architecture, and deployment authorization are not mechanized there.

The worked notebook follows the paper's examples. Figures are generated from those exact example values; rendering converts them to floating-point display coordinates only. Current commands and results are recorded in validation.

10.2 What the older records establish

Historical report (CODA-15). The recovery note describes a 2026-09-01 survey of 865 Lean files, 773 distinct contents, 92 duplicates, 261 Mathlib importers, and 262 cited names. Its recorded scan classified 250 names as declarations, five as mentions, and seven as missing. The note also reports builds from scratch copies whose source copies and six ignored logs were not committed. Those logs are unavailable in the current package; their earlier reported outcomes are summarized with this limitation in historical build records. The source manifest pins the original recovery note, scanner and scan output externally. The scan locates declaration names; it does not establish semantic correspondence or a successful current build. No local custody or replay claim is made for the missing logs or scratch sources.

The source ledgers identify missing or placeholder theorem homes, conflicting status tables, and a declared `anomalon_exists` axiom. The consolidation neither upgrades those claims nor declares that their evidence must still be missing in every later checkout. Their current disposition is historical or open until individually re-established. The original claim IDs remain mapped in the unified ledger; the proof of a narrower statement does not discharge a broader historical claim.

External boundary (CODA-16). The sealed derivation is cited only as `sealed-derivation-v1`, commit `3100c4e`. Its recorded mechanical stopping conditions are not a verdict on all mathematical interpretations. No sealed source or gate implementation is imported into this package. The local elementary proofs were written on the CODA side from the stated definitions. Later research references in the externally pinned design retain their immutable commits and historical status; they are not recertified by this consolidation.

10.3 The remaining programme

Open programme (CODA-18). The source design's N1–N10 and D1–D9 targets remain a dated research agenda, available in its externally pinned §§7–8. The consolidated package makes no new completion claim for them. Their central obligations can be stated as follows:

Obligation	Evidence needed to advance it
Lost or mismatched proof homes	Exact source, build root, theorem statement, axiom report, and claim comparison
A successful faithful representation	The category construction and hom-set injectivity required in §9
Self-hosting and a running instance	An actual encoded run with replayed events and terminal correspondence
Endogenous demand, stake, and selection	A theorem or countermodel in a stated grammar and candidate class
Removal and unprompted transitions	A new event grammar and rechecked preservation laws
Multiple observers	Declared observation maps and query/repair comparisons on a common domain
Limits and continuum structure	Stage maps, topology or metric, existence, compatibility, and relevant convergence proofs
Costs and conserved quantities	Explicit measures with invariant, monotonicity, or transformation laws
General mathematical comparisons	A sound interpretation or other precisely stated model construction
Publication or catalog admission	The applicable author, producer, metadata, and release decisions

This package does not derive Hilbert space, the Born rule, a physical clock, an exhaustive classification of mathematical structures, or an unconditional authorization rule. It does not close the sealed derivation's residual register, establish an intrinsic origin of every new question, or turn finite agreement into semantic closure.

11. Conclusion

CODA's unified contribution here is a sequence of explicit obligations. A supplied distinction precedes an observation; a query descends exactly when it is constant on the observation fibers; a repair result is quantified over its declared family; a transport or global section needs its own compatibility laws; an action needs the permission required by its stated policy. The examples show where each implication succeeds and where it fails. The claim ledger preserves the difference between these results, their implementations, and the programme's open constructions.

Source references and reading map

Key	Source	Role in this paper
S-DES	Historical design (v1.1)	Operational loop, component census, proposed notation, dated research targets

Key	Source	Role in this paper
S-FDN	Foundational manuscript	Witnessed difference, image factorization, menu scope, and shared interfaces
S-DIS	Discipline statement	Author-fixed name, declaration pricing, withdrawals, and faithful-functor target
R-A	A-series recovery	Beginning, observer failures, residual typing, and conditional anomalon claims
R-CS	CS-series recovery	Finite shadows, detector adequacy, image-reverse limits, and bridge evidence
R-F	FDA, FOC, NSA recovery	Repair quotients, verifier and authority boundaries, and nonselection scope
R-M	MUR, OR, discipline recovery	Transport, loss, compatibility, integral example, and target construction
R-Q	Q-series recovery	Quarantined questions, collision barriers, and finite/continuum non-inference
R-L	Lean recovery	Dated source census and limits of historical build custody
P-004	Paper 004	Role Non-Impersonation and Authority Preservation
P-005	Paper 005	The Authority Gap (Verification != Authorization)
P-027	Paper 027	The Authority Firewall: Machine-Checked Separation of Evidence from Authority
P-028	Paper 028	Finite Authority and Non-Terminal Computation
P-029	Paper 029	The Anomalon Boundary: Difference and Structure
P-030	Paper 030	Authority Firewall and Verification Limits

Historical file-and-line references inside these notes may point to an earlier estate. The source manifest distinguishes seventeen byte-verified local sources from twenty-two externally pinned tracked inputs. It does not independently recover every external object cited by those inputs.